Indian Institute of Technology Gandhinagar

MA 104: ODE

Semester-II, Academic Year: 2025-26

Tutorial – 6: Part (a)

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(1) Find a basis of solutions of the ODE

$$(1 - x^2)y'' - xy' + p^2 y = 0.$$

Show that if p is a positive integer, then there exists a polynomial of degree p which satisfies the given ODE.

(2) Express $\sin^{-1} x$ in the form of a power series $\sum_{n=0}^{\infty} a_n x^n$ by solving $y' = (1 - x^2)^{-1/2}$. Use this result to obtain the formula

$$\frac{\pi}{6} = \frac{1}{2} + \frac{1}{2}\frac{1}{3 \cdot 2^2} + \frac{1 \cdot 3}{2 \cdot 4}\frac{1}{5 \cdot 2^4} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\frac{1}{7 \cdot 2^6} + \cdots .$$

(3) Show that for $m < n$ as well as $m - n = 2k + 1$, where k is a non-negative integer,

$$\int_{-1}^{1} x^m P_n(x)\, dx = 0.$$

(4) The *generating function* of a sequence $\{f_n\}_{n \geq 0}$ is the function $G(x, t) = \sum_{n=0}^{\infty} f_n t^n$. Show

$$G(x, t) := \frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n$$

is the generating function of the Legendre polynomials $(P_n(x))_{n \geq 0}$. Using this prove that

(i) $(2n + 1)xP_n(x) = (n + 1)P_{n+1}(x) + nP_{n-1}(x)$,

(ii) $nP_n(x) = xP_n'(x) - P_{n-1}'(x)$,

(5) Show that for $n \neq m$,

$$\int_{-1}^{1} (1 - x^2)P_m'(x)P_n'(x)\, dx = 0.$$

(6) Show that under the transformation

$$y(x) = u(x) \exp\left(-\frac{1}{2}\int p(x)\, dx\right),$$

the equation

$$y'' + p(x)y' + q(x)y = 0$$

reduces to

$$u'' + Q(x)u = 0,$$

where

$$Q(x) = q(x) - \frac{p^2(x)}{4} - \frac{p'(x)}{2}.$$

Use the above method to solve

$$x^2y'' + xy' + \left(x^2 - \frac{1}{4}\right) y = 0, \qquad x > 0.$$

(7) Find a basis of solutions by the Frobenius method.

(a) $xy'' + (2x+1)y' + (x+1)y = 0$

(b) $xy'' + (1-2x)y' + (x-1)y = 0$

(c) $x^2(x^2-1)y'' - x(1+x^2)y' + (1+x^2)y = 0$

(d) $x(x-1)y'' + 2(2x-1)y' + 2y = 0$

(8) Show that between any two consecutive positive zeros of $J_n(x)$ there is precisely one zero of $J_{n+1}(x)$ and vice versa.

(9) Find a general solution in terms of J_ν and $J_{-\nu}$ or indicate when this is not possible. Use the indicated substitutions.

(a) $x^2y'' + xy' + (\lambda^2x^2 - \nu^2)y = 0$, with $\lambda x = z$

(b) $x^2y'' + xy' + \frac{1}{4}(x^2-1)y = 0$, with $x = 2z$

(c) $xy'' + (2\nu+1)y' + xy = 0$, with $y = x^{-\nu}u$

(10) [1]Using recurrence relations, show that $xJ'_{n+1}(x) + (n+1)J_{n+1}(x) = xJ_n(x)$.

(11) Show that the Bessel functions J_ν ($\nu \geq 0$) satisfy

$$\int_0^1 x\, J_\nu(\lambda_m x)\, J_\nu(\lambda_n x)\, dx = \frac{1}{2} J^2_{\nu+1}(\lambda_n)\, \delta_{mn},$$

where λ_n are the positive zeros of J_ν.

[1]For Quiz 2, one may skip Questions 10 and 11.